Here's a project idea to practice Scrum in one day: **"Create a Minimalist Computer Game"**.

Lab Outline: "Scrum Game Jam"
Objective:
Students will practice Scrum by working in a team to design, develop, and present a simple computer game prototype within one day.

Project Description
The team will create a minimalist 2D computer game (e.g., a platformer, puzzle, or arcade-style game). The game should have:

1. A clear goal or win condition.
2. Basic gameplay mechanics (e.g., movement, scoring, or enemies).
3. Simple visual assets (can be placeholders).
4. At least one level or challenge to play.

Roles
Each team member takes on a specific Scrum role:

1. **Product Owner (PO)**
    - Defines the game concept and user stories.
    - Prioritizes the Product Backlog.
    - Ensures the team understands the vision.
2. **Scrum Master (SM)**
    - Facilitates the process.
    - Ensures adherence to Scrum principles.
    - Removes obstacles for the team.
3. **Development Team Members** (everyone else)
    - Collaboratively design, develop, and test the game.
    - Contribute across disciplines (coding, design, testing, etc.).

Schedule (One-Day Sprint)

1. **Sprint Planning (30 minutes)**
    - PO explains the vision and prioritizes the backlog.
    - Team discusses and commits to tasks for the sprint.
2. **Morning Work Session (2 hours)**
    - Develop core mechanics and assets.
3. **Daily Scrum (15 minutes)**
    - Quick stand-up: What was done? What's next? Any blockers?
4. **Afternoon Work Session (2 hours)**
    - Build additional features, test, and refine.
5. **Sprint Review (45 minutes)**
    - Present the game to stakeholders (the instructor or other teams).
    - Gather feedback.
6. **Sprint Retrospective (30 minutes)**
    - Discuss what went well, what didn't, and how to improve.

Tools and Technologies

- **Game Engine**: Use a beginner-friendly engine like Unity (C#) or Godot (GDScript).
- **Task Management**: Trello, Jira, or a physical board for backlog management.
- **Collaboration**: GitHub or another version control system.

Example Game Ideas

1. **Escape the Maze**: A player navigates a maze while avoiding obstacles.
2. **Falling Blocks**: A puzzle game like Tetris.
3. **Alien Invaders**: A simple arcade shooter.
4. **Platform Jumper**: A character jumps between platforms to reach the end.

Assessment Criteria

1. **Teamwork**: How well the team followed Scrum roles and ceremonies.
2. **Execution**: Functionality and playability of the game.
3. **Presentation**: Clear explanation of the game and the Scrum process.
4. **Reflection**: Insights from the retrospective.

Here's an expanded version of the **Product Backlog, Sprint Stages, and Sprint Backlog Items** with more detailed tasks and roles for each stage.

Expanded Product Backlog

1. **Basic Player Movement (Priority: High)**
    - Task: Implement keyboard input to allow movement in four directions (up, down, left, right).
    - Acceptance Criteria: The player moves smoothly and stops at walls or maze boundaries.
2. **Maze Layout (Priority: High)**
    - Task: Design a static maze using a grid system.
    - Acceptance Criteria: The maze includes paths and walls that guide the player toward a goal.
3. **Win Condition (Priority: High)**
    - Task: Create a designated goal area.
    - Acceptance Criteria: Reaching the goal ends the game and displays a "You Win!" message.
4. **Obstacle Mechanics (Priority: Medium)**
    - Task: Add static or moving obstacles that block the player's path or cause them to lose.
    - Acceptance Criteria: Colliding with an obstacle ends the game and displays a "Game Over" screen.
5. **Scoring System (Priority: Medium)**
    - Task: Implement a scoring system based on time or collectible items.
    - Acceptance Criteria: A score is displayed on-screen and updated dynamically during gameplay.
6. **Visual Design (Priority: Low)**
    - Task: Create basic graphics for the maze, player, and obstacles.
    - Acceptance Criteria: Graphics are simple but distinguishable (e.g., player as a circle, walls as squares).
7. **Sound Effects and Music (Priority: Low)**
    - Task: Add sound effects (e.g., for movement or collision) and background music.
    - Acceptance Criteria: Sounds play at appropriate times, and music loops correctly.
8. **Game Start Menu (Priority: Medium)**
    - Task: Create a menu with "Start Game" and "Exit" buttons.
    - Acceptance Criteria: The player can start or exit the game from the menu.
9. **Game Over Screen (Priority: Medium)**
    - Task: Display a "Game Over" screen with an option to retry.
    - Acceptance Criteria: The screen appears after a loss and allows restarting the game.

Expanded Sprint Stages and Sprint Backlog Items

Sprint 1: Build the Core Gameplay

Goal: Deliver a functional prototype with player movement and a basic maze layout.

Sprint Backlog Tasks:

1. **Implement Player Movement**
    - Assign: Developer 1 (Coding).
    - Sub-tasks:
        - Write input handling for arrow keys.
        - Create movement logic tied to the grid system.
        - Test smooth movement and ensure no overlap with walls.
2. **Design and Implement a Basic Maze Layout**

    - Assign: Developer 2 (Coding + Design).
    - Sub-tasks:
        - Create a grid structure for the maze.
        - Define walls and paths.
        - Hardcode a simple maze design for testing.
3. **Add the Goal Area and Win Condition**
    - Assign: Developer 3 (Coding).
    - Sub-tasks:
        - Define the goal area (e.g., a green square).
        - Detect player reaching the goal.
        - Display a "You Win!" message when the goal is reached.
4. **Basic Collision Detection**
    - Assign: Developer 4 (Coding).
    - Sub-tasks:
        - Implement collision detection with walls.
        - Ensure the player cannot move through walls.

Deliverable: A playable prototype where the player can move through a maze and win by reaching the goal.

Sprint 2: Add Challenge and Feedback
Goal: Introduce obstacles, a scoring system, and basic game feedback.
Sprint Backlog Tasks:

1. **Implement Moving Obstacles**
    - Assign: Developer 1 (Coding).
    - Sub-tasks:
        - Create enemy or obstacle objects.
        - Program movement patterns (e.g., back-and-forth).
        - Test collision between the player and obstacles.
2. **Add a Scoring System**
    - Assign: Developer 2 (Coding).
    - Sub-tasks:
        - Create a timer-based scoring mechanism.
        - Display the score on-screen.
        - Add collectible items that increase the score.
3. **Create a Game Over Screen**
    - Assign: Developer 3 (Coding).
    - Sub-tasks:
        - Display a "Game Over" message on losing.
        - Add a "Retry" button to restart the game.
4. **Refine Collision Mechanics**
    - Assign: Developer 4 (Coding).
    - Sub-tasks:
        - Ensure all collision interactions are consistent.
        - Handle cases where the player collides with multiple objects simultaneously.

Deliverable: A more engaging prototype with obstacles, scoring, and clear feedback for the player.

Sprint 3: Enhance User Experience
Goal: Improve visuals, sound, and user interface for a polished game experience.
Sprint Backlog Tasks:

1. **Add Visual Design Elements**
    - Assign: Developer 1 (Design).

    - Sub-tasks:
        - Create basic sprites for the player, obstacles, and goal.
        - Add textures or colours to the maze.
2. **Include Sound Effects and Music**
    - Assign: Developer 2 (Audio).
    - Sub-tasks:
        - Find free-to-use sound effects and music.
        - Add sound for movement, collision, and reaching the goal.
3. **Create a Start Menu**
    - Assign: Developer 3 (UI).
    - Sub-tasks:
        - Design a simple menu with "Start Game" and "Exit" options.
        - Link menu buttons to corresponding actions.
4. **Finalize the Game Over Screen**
    - Assign: Developer 4 (UI).
    - Sub-tasks:
        - Polish the "Game Over" screen design.
        - Ensure the "Retry" button resets the game state correctly.

Deliverable: A polished game prototype with improved visuals, sound, and user interface.

Optional Additional Tasks (If Time Allows)

- Add dynamic difficulty (e.g., faster obstacles over time).
- Implement multiple levels with increasing complexity.
- Allow the player to customize the appearance of the maze or character.

Team Role Assignments

- **Scrum Master**: Facilitates all Scrum ceremonies, ensures progress, and resolves blockers.
- **Product Owner**: Focuses on prioritizing tasks and maintaining the product vision.
- **Developers**: Collaborate to implement tasks as per the sprint backlog.

Let me know if you'd like further breakdowns, such as code structure or example scripts!